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# Python Fundamentals – Knowledge Test

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Duration: 30 Minutes

Marks: 30

Instructions:

1. All questions are multiple choice.
2. Choose the correct option.
3. Each question carries 1 mark.

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## Core Python Concepts

Q1

In Python, which of the following is a requirement for defining a block of code, such as the body of a function or a loop?

- A. Ending every line with a semicolon
- B. Beginning the block with a `begin` keyword
- C. Consistent indentation, typically using 4 spaces
- D. Enclosing the code block within curly braces

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Q2

What is the resulting value and data type of the operation `15 // 4` in Python?

- A. 3.75 (Float)
- B. 4 (Integer)
- C. 3 (Float)
- D. 3 (Integer)

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Q3

If you use the `input()` function to collect a user's age, what must you do before performing a mathematical comparison like `age > 18`?

- A. Nothing, Python automatically detects the numeric type
- B. Use the `.strip()` method to remove quotes

- C. Store it in a tuple to make it a number
  - D. Convert the input to an integer using `int()`
- 

#### Q4

When validating an email address using the `re` module, which function is best suited to check if the string matches a pattern from the very beginning?

- A. `re.split()`
  - B. `re.findall()`
  - C. `re.match()`
  - D. `re.search()`
- 

#### Q5

A student wants to create a collection of Chennai Metro station codes where no duplicates are allowed and the order is not important. Which data structure should they use?

- A. Set
  - B. Tuple
  - C. Dictionary
  - D. List
- 

#### Q6

Which string method would you use to change the string `"naukri learning"` into `"Naukri Learning"`?

- A. `.upper()`
  - B. `.replace()`
  - C. `.capitalize()`
  - D. `.title()`
- 

#### Q7

What is the result of the function call `chr(65)` in Python?

- A. Error
  - B. 65
  - C. 'A'
  - D. 'a'
- 

#### Q8

Which of the following describes a "Strong Number"?

- A. A number that equals the sum of the factorials of its digits
  - B. A number divisible only by 1 and itself
  - C. A number that reads the same forwards and backwards
  - D. A number that is the sum of its proper divisors
- 

### Q9

When working with files, what is the advantage of using the `with open(...)` statement over manually calling `file.close()`?

- A. It automatically closes the file even if an error occurs
  - B. It makes the file read-only
  - C. It allows multiple programs to edit the file simultaneously
  - D. It speeds up file reading by 50%
- 

### Q10

Which comparison operator is used to check if two values are NOT equal?

- A. `<>`
  - B. `!=`
  - C. `not ==`
  - D. `==`
- 

### Q11

How does Python handle variable names like `student_name` and `Student_Name`?

- A. They are treated as two different variables
  - B. Python merges them automatically
  - C. They cause a naming conflict
  - D. The second one is invalid
- 

### Q12

What is the output of the following code?

```
print(type(10 / 2))
```

- A. `<class 'int'>`
  - B. `<class 'float'>`
  - C. `<class 'str'>`
  - D. Error
-

## Q13

Which of the following correctly converts user input into a float?

- A. `float(input())`
  - B. `input(float())`
  - C. `int(input())`
  - D. `str(input())`
- 

## Q14

Which data structure is best suited for storing key-value pairs?

- A. List
  - B. Tuple
  - C. Dictionary
  - D. Set
- 

## Q15

What is the output of:

```
print("Python".lower())
```

- A. PYTHON
  - B. python
  - C. Python
  - D. Error
- 

## Q16

Which function is used to search for a pattern anywhere in a string using regex?

- A. `re.match()`
  - B. `re.search()`
  - C. `re.split()`
  - D. `re.compile()`
- 

## Q17

What is the output of:

```
x = "10"  
y = int(x)  
print(y + 5)
```

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- A. 105
  - B. 15
  - C. Error
  - D. "105"
- 

### Q18

Which of the following is NOT a valid Python variable name?

- A. student1
  - B. \_name
  - C. 1student
  - D. student\_name
- 

### Q19

What is the output of:

```
print(len("Learning"))
```

- A. 7
  - B. 8
  - C. 9
  - D. Error
- 

### Q20

Which statement is true about Python sets?

- A. They allow duplicate values
  - B. They maintain insertion order strictly
  - C. They store unique elements
  - D. They only store integers
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